

EE 5239 Nonlinear Optimization: Topics by Lectures

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In this document, I will keep an update on the important topics by each lecture (up to Lecture 6). I hope that this document can serve as a basis in which HWs and Mid-term can be prepared. It also outlines which are the essential materials, which are more advanced materials.

1 Unconstrained Optimization

Note: “not required” here means not required in Mid-Term.

- Lecture 1 (week 1-2)
 1. Definition of unconstrained problems
 2. Existence of optimal solution
 3. Necessary and sufficient conditions for local min
 4. Know how to use these conditions
 5. Convexity (various ways of determining the convexity of a function)
 6. Why all local min are global min for convex problems
 7. **Readings:** Book Sec 1.1 Appendix A, Appendix B1. (V2, V3)
- Lecture 2 (week 2-3)
 1. The basic steps of gradient methods, how to choose stepsizes
 2. Different way of choosing stepsizes/directions
 3. Know what the Descent Lemma means, and how to use it
 4. Understand how to estimate the descent of steepest descent method with constant step-size, i.e., how to obtain the following in Lecture 2b.
$$f(x^{r+1}) \leq f(x^r) - \frac{1}{2L} \|\nabla f(x^r)\|^2 \tag{1}$$
 5. Understand the analysis of steepest gradient descent algorithm with $1/L$ stepsize
 6. General analysis of convergence
 7. Strong convexity, and definition of condition number;
 8. Understand the convergence rate analysis steps; also know how gradient type algorithms depends on condition number
 9. **Readings:** Book Sec 1.2 -1.3, Appendix A, Appendix B1. (V2, V3).
- Lecture 3

1. Know what the conjugate direction and conjugate gradient methods are; know the definition of conjugate directions, and the relationship between Q conjugate vectors, linearly independent vectors, and orthogonal vectors
2. Know how scaled steepest descent works in theory, and how to pick appropriate scaling matrices
3. Know the steps of incremental method; why it will diverge with constant stepsize?
4. Know the steps of the coordinate descent method (2 different versions, where each block performs exact minimization or gradient step)
5. How to apply the previous two methods to the least square regression problem

- Lecture 4

1. Understand the K-means problem formulation
2. Know how to apply coordinate descent to K-means (know how to derive each step), and be able to write down the steps
3. Understand how to implement the Perceptron algorithm, and know how to implement the algorithm for a classification problem
4. Know the convergence statement of Perceptron
5. Know the general proof idea of the convergence of Perceptron

- Lecture 5

1. Understand what causes different practical behavior for algorithm
2. Understand the basics behind logistic regression
3. Understand the difference between *strong* convexity and *strict* convexity, and the implication of the resulting gradient descent based algorithms
4. From optimization perspective, understand why data set 1 and data set 2 will yield different behavior for GD algorithm
5. High dimensional case not required

- Lecture 6a

1. Appreciate the difference between constrained and unconstrained problem
2. Understand the problem we are analyzing, i.e., minimizing an objective function, over a *convex* set
3. Can write down the optimality condition; understand its geometric meaning
4. Know how to solve the simple scalar quadratic problem with bound constraint (page 15)
5. Know how to characterize solutions with bound constraints (vector case, page 16)
6. Understand the projection operator; connecting its defining optimization problem with its geometric representation
7. Know how to compute simple projections, such as for the following sets:

$$a \leq x \leq b, \|x\| \leq 1, \|x\| = 1. \quad (2)$$

8. Understand the property in page 21 of the slides.

9. Understand Example 3.1.2 (V3)/2.1.2(V2), (Optimization over a Simplex)
10. Understand Example 3.1.1 (V3)/2.1.1(V2), (Optimization subject to Bounds)

- Lecture 6b

1. Definition of feasible directions
2. Feasible direction method, and its relationship with gradient descent
3. Gradient projection methods, and two special cases; Special case II is more popular;
4. Convergence result statement; Prop 3.3.1, Prop 3.3.2 (V3), or Prop 2.3.1, Prop. 2.3.2 (V2);
5. How to perform projection on nonnegative set
6. Know the convergence rate result (prove step is not required in exam)
7. Know the Frank-Wolfe step, and understand geometrics what it is doing; proof is not required;